

SECTION 13: COOLING TOWER SOP

Purpose: Maintain safe and efficient operation of cooling towers to ensure proper heat rejection from chilled water systems through heat transfer to the condenser water lines. Prevent scaling, biological growth, and structural degradation.

Daily Operational Procedures:

Inspect tower basin for debris, oil sheen, and water level.

Check operation of float valve and makeup water feed.

Measure basin conductivity and verify chemical dosing.

Record ambient and return air temperatures.

Verify fan operation and inspect belts or drive motors.

Confirm louvers and drift eliminators are clean and secure.

Weekly Maintenance Tasks:

Clean strainers and inspect screens for clogs.

Inspect fan bearings and lubricate if necessary.

Verify bleed-off valve is functioning.

Check for corrosion, scaling, and biological growth in basin.

Review and calibrate conductivity sensors.

Safety Notes:

Wear gloves and goggles when handling treatment chemicals.

Use fall protection when accessing elevated platforms or fans.

Applicable Guidelines:

ASHRAE Guideline 12-2020

OSHA 1910 Subpart D (Walking-Working Surfaces)

ANSI/ASHRAE Standard 188 (Legionella control)